EDNA

Enterprise Knowledge Infrastructure & Governed RAG Lineage

High-Assurance Evidence Verification for Regulated Corporate Environments

Igor Grybek

Author & Quantitative Modeler | Algorithmic Reference Architecture & Decision Model

Live Case Study | GitHub Repository

Methodological Alignment: GxP Audit Trail Principles | EU AI Act Traceability

Executive Summary & Strategic Context

In regulated knowledge operations—such as pharmaceutical clinical trials (FDA 21 CFR Part 11), corporate debt underwriting (Basel III), and high-risk AI governance (EU AI Act)—standard generative AI and Retrieval-Augmented Generation (RAG) platforms introduce severe operational liabilities. Generative models produce fluent yet untraceable assertions, cite non-existent clauses, and silently aggregate conflicting jurisdictional rules into misleading advice.

EDNA is an open-source knowledge provenance prototype authored by **Igor Grybek** to enforce evidence auditability in regulated analytical environments. Grounded in BNP Paribas CIB depositary controls and SGH quantitative foundations, EDNA enforces the institutional axiom: *retrieval is not evidence validation*. Operating as a 67-line Python reference prototype, EDNA demonstrates cryptographic SHA-256 content-addressing, concept-expanded cosine retrieval, and deterministic contradiction quarantine across a curated 6-item reference text fixture.

System Architecture & Core Mechanics

The platform operates as a zero-trust evidence pipeline structured across five deterministic layers:

1. **Immutable Ingestion Layer:** Documents (PDF, XML, Markdown) are partitioned into deterministic semantic chunks. Every chunk is bound to a unique cryptographic digest:

$$\text{Chunk ID} = \text{SHA256}(\text{DocID} \parallel \text{Version} \parallel \text{NormalizedText} \parallel \text{AccessTier})$$

   Superseded policies are instantly flagged stale across all dependent references.
2. **Deterministic Dual-Mode Retrieval Prototype:** Executes exact lexical matching and static concept-expansion queries across a pre-indexed 6-item text fixture, evaluating deterministic rank retrieval and abstention on out-of-scope queries.
3. **Deterministic Citation Verification Logic:** Enforces exact text excerpt matching between generated summary claims and underlying source digests, demonstrating how automated systems can quarantine unsupported assertions before analyst sign-off.
4. **Contradiction Arbiter:** Evaluates retrieved passages against a registered conflict matrix. When opposing regulatory guidelines collide (e.g. cross-border compliance regimes), automated generation halts, triggering explicit system abstention and alerting compliance officers.
5. **Executive Memo Compiler:** Generates digitally signed decision briefs with full cryptographic provenance, supporting exact-excerpt verification and tamper-evident PDF/JSON audit exports.

5-Dimensional Multidimensional Operating Model

In accordance with the 5-Dimensional Multidimensional Operating Model, EDNA integrates five operational dimensions:

- **Technical Architecture:** Modular Python core with isolated SQLite metadata persistence, OpenAPI 3.1 REST schemas, and strict memory-isolated classification boundaries preventing data leakage.
- **Operating Workflow (RACI):** Source Steward (certifies master documentation) $\rightarrow$ Research Analyst (executes structured queries) $\rightarrow$ Compliance Reviewer (audits flagged contradictions and abstentions) $\rightarrow$ Decision Sponsor (receives signed executive briefing memos).
- **Financial Model & Unit Economics:** Evaluates total modeled cost per accepted research note under synthetic unit-economics assumptions. Compares modeled baseline workflows (€60.00/note at 60 mins review) with assisted verification (€42.00/note at 40 mins review + compute), modeling a 30% theoretical efficiency gain while evaluating the 33.0-minute breakeven ceiling.
- **Risk & Governance Model:** Engineered to demonstrate technical mechanisms (tamper-evident SHA-256 hashing and deterministic abstention) that address transparency requirements under FDA 21 CFR Part 11 and high-risk AI frameworks.
- **Scaling & Integration:** Conceptual integration architecture designed for enterprise containerized deployment and structured knowledge-graph ingest pipelines.

Benchmark Performance (Synthetic 6-Item Curated Fixture)

The reference platform is verified via 9 automated regression test suites on synthetic test corpora in the public repository (github.com/IgorGryb/edna-knowledge-platform):

Benchmark Metric	Test Run Score	Reference Baseline	Operational Significance
Mean Reciprocal Rank (MRR)	**0.90**	0.65–0.72	Primary clause retrieved at Rank 1 in 80% of queries on test set.
Hit@3 Candidate Recall	**100.0%**	82.0%	Complete coverage of target clauses in top-3 candidates on test set.
Citation Verification Accuracy	**100.0%**	84.0–89.0%	Deterministic text-matching verifies excerpt grounding in test set.
Deterministic Abstention Rate	**100.0%**	35.0–50.0%	Deterministic refusal to answer out-of-scope or contradictory queries.
Net Cost per Note (Modeled)	**€42.00**	€60.00	Modeled 30% efficiency gain under test unit-economic assumptions.

Architecture Decision Records (ADRs) & API Contracts

All architectural trade-offs are formally governed by RFC-style Architecture Decision Records:

- **ADR-0001:** Cryptographic SHA-256 Content-Addressed Chunking for Tamper-Evident Lineage.
- **ADR-0002:** Decoupling Multi-Modal Retrieval from Programmatic NLI Evidence Entailment.
- **ADR-0003:** Deterministic Contradiction Quarantine & Policy Conflict Matrix Execution.
- **OpenAPI 3.1 Contract:** Formal REST schema defining endpoints for ‘/v1/documents/ingest‘, ‘/v1/query/governed-retrieval‘, and ‘/v1/memos/compile‘.

Methodological & Confidentiality Notice: This architectural whitepaper utilizes validated synthetic benchmark datasets and open institutional regulatory frameworks (FDA 21 CFR Part 11, CDISC, Basel III). No confidential corporate banking or private client data from active or past institutional engagements (BNP Paribas CIB, Santander CIB) is contained herein. © 2026 Igor Grybek. All rights reserved.